

PHYS3071

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Introduction

About These Notes

These notes accompany the PHYS3071 Astrophysics course in Hong Kong University of Science and Technology (HKUST). They are intended as a self-contained supplement to the lectures, providing both conceptual grounding and mathematical rigour at a level appropriate for readers with a background in basic physics, basic linear algebra, and basic calculus.

The material covers different astrophysical topics, including but not limited to the formation and evolution of stars, the type and spectrum of stars. Each topic is structured to serve different learning objectives—from building physical intuition to solving examination-style problems and exploring advanced derivations.

These notes do not replace lecture slides or the recommended textbooks. They are best used alongside them as a structured study companion.

If you find any errors or have suggestions for improvement, please feel free to contact me at yhip@connect.ust.hk. You can check the latest version of the notes on GitHub: <https://github.com/AuTomatoc-Ti/phys3071-Astrophysics-notes/blob/main/main.pdf>.

Document Conventions

Block Components

Each topic is presented using a consistent **5-component block structure** designed to support multiple learning styles:

PHYSICS INTUITION

Purpose: Builds conceptual understanding through plain language explanations.

This section provides the *motivation* behind the physics concept, explores its *historical development*, and offers *intuitive explanations* that help you grasp what the theory is really about before diving into equations.

Use this when: You're first encountering a new concept or need to refresh the "big picture" before tackling mathematics.

THEORY

Purpose: Provides rigorous mathematical treatment of the concept. Here you'll find detailed *derivations*, *fundamental assumptions*, key *equations*, and connections between *mathematics and physical intuition*. This is the technical foundation required for exams and deeper problem-solving.

Use this when: You need to understand how to apply formulas, prepare for exams, or work through problems rigorously.

QUESTION EXAMPLES

Purpose: Demonstrates practical application through worked problems.

Each question example mimics exam-style problems and includes complete *solution walkthroughs*. These help you bridge the gap between theory and problem-solving, and expose common pitfalls.

Use this when: Practicing problem-solving, preparing for assessments, or checking if you can apply the formulas correctly.

FOUNDATION

Purpose: Supplies background knowledge and detailed proof steps that underpin the main content.

This section presents *prerequisite mathematics or physics*, *step-by-step derivations* that are too detailed to include in the Theory block, and *intermediate results* needed for a thorough understanding. Reading this is optional for the exam but essential for genuine mastery.

Use this when: A derivation in Theory feels too abrupt, you want to verify an intermediate step, or you need to solidify prerequisite concepts.

EXTRA

Purpose: Provides supplementary material that extends *beyond* the course syllabus.

This section offers *interesting asides*, *historical context*, *connections to current research*, and citations to deeper references. The content here is intentionally out-of-scope for the course examination but enriches the broader picture of each topic.

Use this when: You're curious to explore beyond the syllabus, or when a topic connects to your personal research interests.

Suggested Reading Paths:

1. **Exam Preparation:** *Physics Intuition* → *Theory* → *Question Examples*
2. **Deep Learning:** *Physics Intuition* → *Foundation* → *Theory* → *Question Examples* → *Extra*
3. **Quick Review:** *Theory* → *Question Examples*

Special Annotations

In addition to the five block components above, two inline annotations appear throughout the text to flag specific types of information:

IMPORTANT NOTE

Purpose: Alerts you to a result, condition, or sign convention that is easy to get wrong or is frequently tested.

An Important Note signals a *common misconception*, a *critical sign*, a *non-obvious condition* (e.g. a constraint that must hold for an equation to be valid), or an *exam trap* worth memorising explicitly.

Treat these as mandatory reading regardless of which reading path you follow.

REMARK

Purpose: Highlights a useful observation, connection, or naming convention.

A Remark draws attention to a *conceptual link* between results, an *elegant reformulation*, a *naming convention* (e.g. why a quantity carries its particular name), or a *minor but clarifying point* that does not rise to the level of an Important Note.

Use this when: Looking for a deeper understanding of why an equation takes the form it does, or for useful mnemonics.

Mathematical Conventions

Key symbols.

Symbol	Meaning	Typical units
$M_{\odot}$	Solar mass	kg
$R_{\odot}$	Solar radius	m
$L_{\odot}$	Solar luminosity	W
T_{eff}	Effective temperature	K
μ	Mean molecular weight	—
ρ	Mass density	kg m^{-3}

REMARK

Unless otherwise stated, stellar quantities are quoted in solar-normalised form (for example $M/M_{\odot}$ and $L/L_{\odot}$), which makes scaling relations easier to compare across different stellar masses.

differential notation. We use $\dot{x} \equiv \frac{dx}{dt}$ for time derivatives of some variables $x(t)$.